



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 1 Multiple Choice

8873/01
20 September 2018
1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in soft pencil on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator expected, where appropriate.

This document consists of **14** printed pages and **2** blank pages.

- 1 Use of the *Data Booklet* is relevant to this question.

What is the number of atoms in 800 cm^3 of hydrogen gas under room temperature conditions?

- A 2.0×10^{22}
 B 4.0×10^{22}
 C 2.0×10^{25}
 D 4.0×10^{25}

- 2 Use of the *Data Booklet* is relevant to this question.

The relative abundances of the isotopes of a sample of carbon are shown in the table below.

relative isotopic mass	12	13	14
relative abundance	100	1.08	0.01

What is the relative atomic mass of carbon in this sample?

- A 12.00 B 12.01 C 12.10 D 12.15

- 3 Four substances **E**, **F**, **G** and **H** have physical properties as shown.

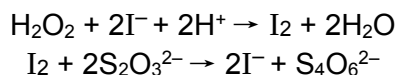
substance	melting point/ °C	boiling point/ °C	electrical conductivity	
			of solid	of liquid
E	17	45	Poor	Poor
F	64	759	Good	Good
G	1132	1950	Poor	Good
H	3550	3825	Good	Unknown

What could be the identities of **E**, **F**, **G** and **H**?

	E	F	G	H
A	PCl_5	SO_3	Na_2O	Al_2O_3
B	PCl_5	Al_2O_3	K	Na_2O
C	SO_3	PCl_5	C (graphite)	Na_2O
D	SO_3	K	Na_2O	C (graphite)

- 4 10 cm³ of hydrogen peroxide was diluted to 250 cm³ using a standard flask. 25.0 cm³ of the diluted solution was reacted with 10 cm³ of concentrated potassium iodide in the presence of dilute sulfuric acid. The resulting mixture was then titrated with 0.10 mol dm⁻³ sodium thiosulfate. It was found that 21.00 cm³ of thiosulfate was required to reach end-point.

The following shows the reactions described above.



What is the initial concentration of the hydrogen peroxide used?

- A 0.04 mol dm⁻³
 B 0.08 mol dm⁻³
 C 1.05 mol dm⁻³
 D 2.10 mol dm⁻³
- 5 Which of the following is a redox reaction?
- A $\text{NaOH} + \text{HBr} \rightarrow \text{NaBr} + \text{H}_2\text{O}$
 B $\text{FeCl}_3 + 6\text{H}_2\text{O} \rightarrow [\text{Fe}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
 C $\text{H}_2\text{SO}_4 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}$
 D $\text{Cu} + \text{NH}_3 + \text{O}_2 + \text{H}_2\text{O} \rightarrow [\text{Cu}(\text{NH}_3)_4](\text{OH})_2$
- 6 Which of the following statements is **untrue** about the second ionisation energies of Period 2 elements?
- A The general trend is similar to that of first ionisation energies.
 B Increase in shielding effect outweighs the increase in nuclear charge.
 C The anomalies lie between group 13 and 14, and between 16 and 17 elements.
 D There is increasing electrostatic forces of attraction between the nucleus and valance electrons.

- 7 When ${}^4\text{He}$ nuclei was pass through an electric field, it was deflected 4° .

Which of the following ions would be deflected half as much as the ${}^4\text{He}$ nuclei in the opposite direction?

- A C^{3-}
 - B N^{3-}
 - C Na^+
 - D Mg^{2+}
- 8 Which of the following statements is true for Al^{3+} ?
- A There are no d orbitals occupied.
 - B There are a total of five subshells occupied.
 - C There are a total of three s orbitals occupied.
 - D The last electron removed to obtain Al^{3+} came from the 3p orbital.
- 9 A chloride and an oxide of the elements in the third period of the Periodic Table were dissolved in two separate portions of water to form aqueous solutions.

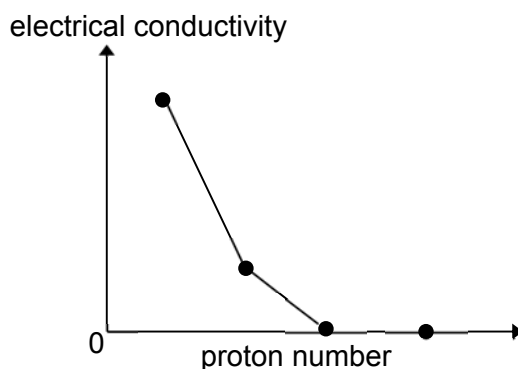
Both of the resulting solutions could dissolve Al_2O_3 but only one of the two could dissolve P_4O_{10} .

Which of the following pairs could be the chloride and the oxide used?

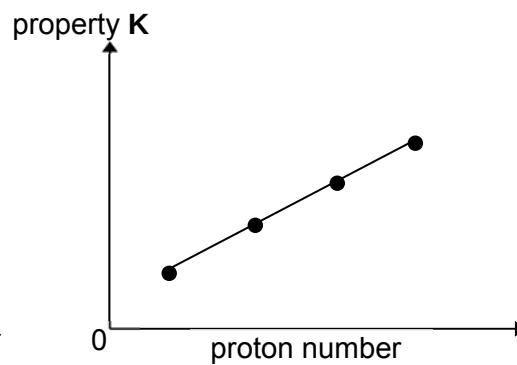
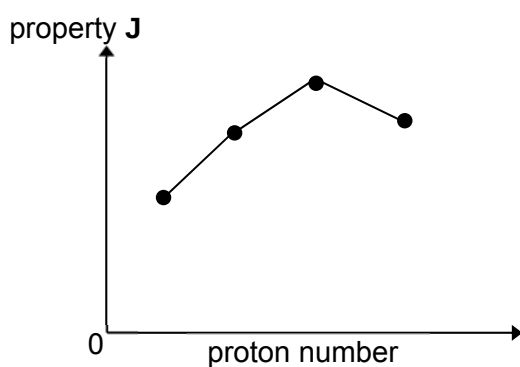
- | | | |
|---|-----------------|-----------------------|
| 1 | NaCl | SO_3 |
| 2 | MgCl_2 | MgO |
| 3 | SiCl_4 | Na_2O |

- A 2 only
- B 1 and 3
- C 2 and 3
- D 1, 2 and 3

- 10 The diagram represents the melting points of four consecutive elements in the third period of the Periodic Table.



The sketches below represent another two properties of the elements.



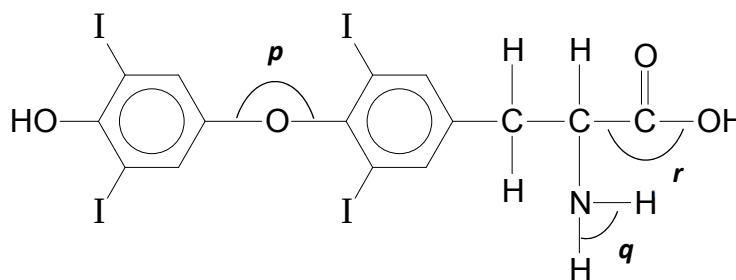
What are properties **J** and **K**?

	property J	property K
A	first ionisation energy	electronegativity
B	number of valence electrons	melting point
C	ionic radius	nuclear charge
D	boiling point	atomic radius

11 Which one of the following **cannot** be explained using hydrogen bonding?

- A CH_3CHO has a higher boiling point than C_2H_6 .
- B CH_3COOH has a M_r of 120 in benzene.
- C HF has a higher boiling point than HCl.
- D CH_3COCH_3 is miscible with water.

12 The thyroid gland concentrates iodine and uses it to produce thyroxine, which is a hormone that controls the metabolic rate.

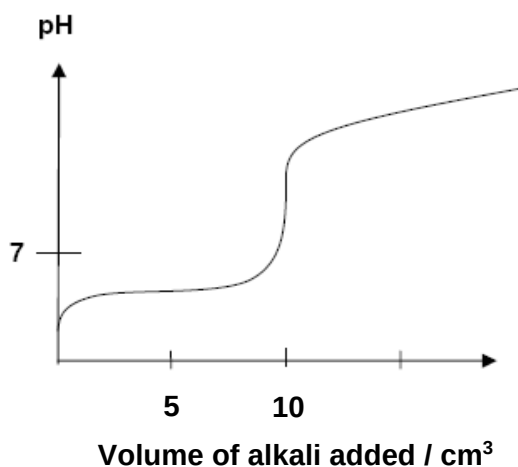


Thyroxine

What are the values of the bond angles *p*, *q* and *r*?

	<i>p</i>	<i>q</i>	<i>r</i>
A	105°	107°	180°
B	105°	107°	120°
C	109°	120°	180°
D	109°	120°	120°

- 13 The following graph shows the changes in pH of 20.0 cm³ of 1.0 mol dm⁻³ CH₃COOH when excess 1.0 mol dm⁻³ alkali solution is added gradually.



Which one of the following alkali solution with a suitable indicator could have resulted in the graph below?

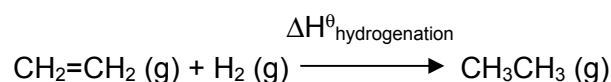
Alkali	Indicator
A NaOH	Methyl orange
B NaOH	Phenolphthalein
C Ba(OH) ₂	Methyl orange
D Ba(OH) ₂	Phenolphthalein

- 14 Which of these mixtures will form a buffer solution?

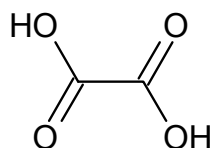
- A 20 cm³ sodium hydroxide and 10 cm³ sulfuric acid of the same concentration
- B Equal volumes of 1.0 mol dm⁻³ sodium hydroxide and 2.0 mol dm⁻³ ammonia
- C Equal volumes of 1.0 mol dm⁻³ ethanoic acid and 1.0 mol dm⁻³ ammonia
- D 10 cm³ ethanoic acid and 20 cm³ sodium ethanoate of the same concentration

- 15 Given the following data, determine the enthalpy of hydrogenation of ethene, $\text{CH}_2=\text{CH}_2$ to ethane, CH_3CH_3 .

compound	enthalpy change of combustion ΔH_c^θ
$\text{CH}_2=\text{CH}_2$ (g)	-1411
H_2 (g)	-286
CH_3CH_3 (g)	-1560



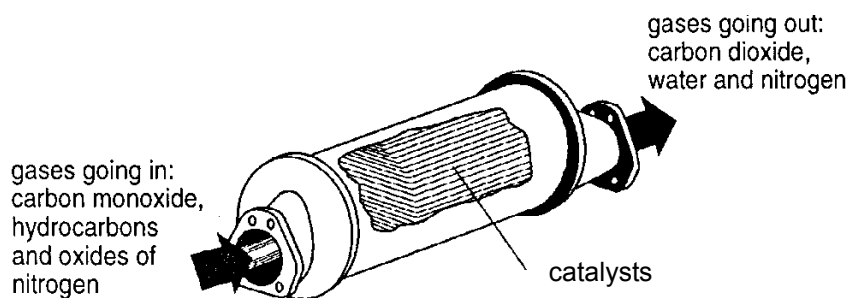
- A +137 kJ mol^{-1}
 B -137 kJ mol^{-1}
 C +1084 kJ mol^{-1}
 D -1084 kJ mol^{-1}
- 16 Which of these statements account for the lower melting point of aluminium oxide as compared to magnesium oxide?
- A Aluminium oxide is a simple molecular structure.
 B Aluminium oxide has weaker intermolecular forces of attraction.
 C Aluminium oxide has a greater covalent character than magnesium oxide.
 D Aluminium oxide has a larger magnitude of lattice energy than magnesium oxide.
- 17 Which statement best explains the reason why the reaction between 10 cm^3 of 0.1 mol dm^{-3} sodium hydroxide with 5 cm^3 of oxalic acid 0.1 mol dm^{-3} is more exothermic than between sodium hydroxide with hydrochloric acid of the same respective volumes and concentrations?



Oxalic acid

- A Some energy is required to further dissociate the weak acid.
 B More molecules of water are produced in the first reaction.
 C Both acids are limiting reagents.
 D Heat loss to the surroundings.

- 18** The diagram shows the structure of a catalytic converter as fitted into the exhaust system of a car. It also describes the substances involved in the exhaust gases.

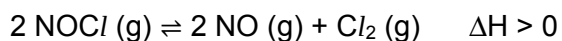


Which processes take place in the catalytic converter?

- 1** Platinum and cobalt catalyse redox reactions.
- 2** Carbon monoxide and oxides of nitrogen react together.
- 3** Carbon monoxide and hydrocarbons react together.
- 4** Hydrocarbons and oxides of nitrogen react together.

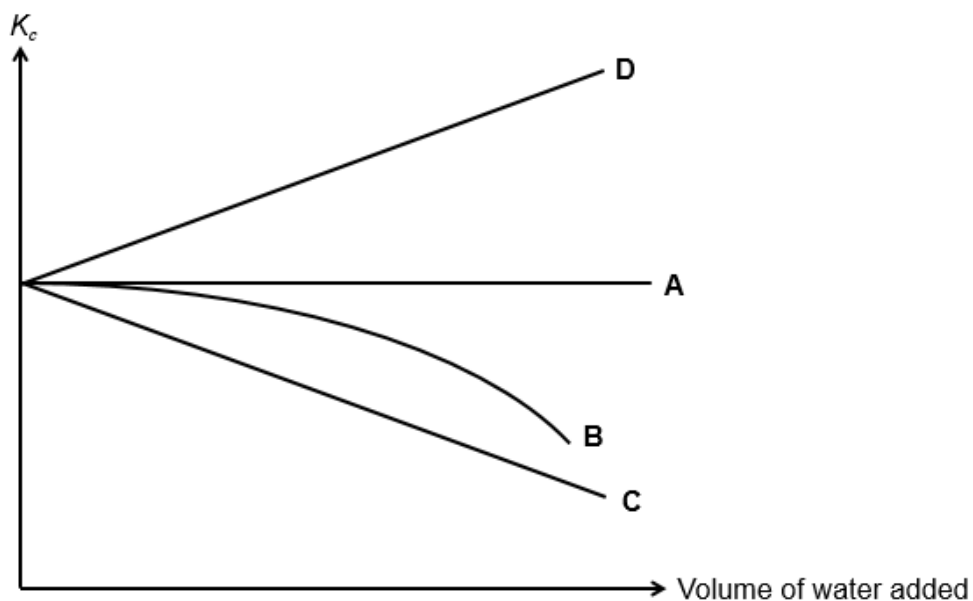
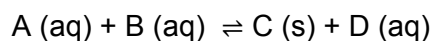
- A** 1 and 2
B 2 and 3
C 2 and 4
D 1 and 3

- 21 At 35°C, $K_c = 1.6 \times 10^{-5} \text{ mol dm}^{-3}$ for the following reaction



Which one of the following statements is correct?

- A $[\text{NO}] < [\text{NOCl}]$ at equilibrium
 - B K_c will decrease when temperature increases.
 - C Addition of catalyst will cause more Cl_2 to be formed.
 - D Position of equilibrium will shift to the right when total pressure of the system increases.
- 22 Which of these graphs is correct for the equilibrium shown below?

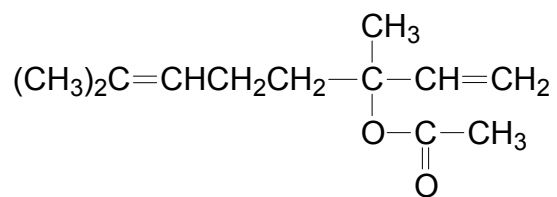


23 Which of the following compounds on reaction will produce ethanoic acid?

- 1 $\text{CH}_3\text{CH}=\text{CH}_2$ and steam
- 2 $\text{CH}_3\text{COOCH}_3$ and dilute HCl , reflux
- 3 $\text{CH}_3\text{CH}_2\text{CHO}$ and acidified KMnO_4 , reflux
- 4 $\text{CH}_3\text{CH}_2\text{OH}$ and acidified $\text{K}_2\text{Cr}_2\text{O}_7$, reflux

- A 1 and 2
 B 2 and 3
 C 3 and 4
 D 2 and 4

24 Linalyl acetate is a naturally-occurring compound and it is a principal component of the essential oils of lavender.

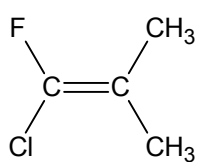


Linalyl acetate

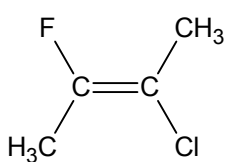
Which one of the statements is **not** true about linalyl acetate?

- A It is a condensation polymer.
 B It cannot exhibit cis-trans isomerism.
 C It reacts with hot sulfuric acid.
 D It reacts with 2 mol of hydrogen gas.

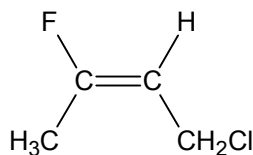
25 Which of the molecular structures below will have the smallest overall dipole?



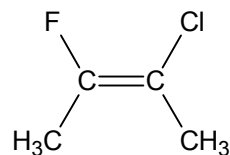
A



B



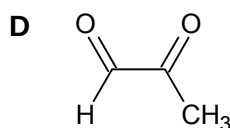
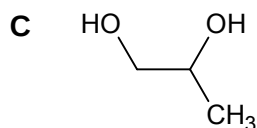
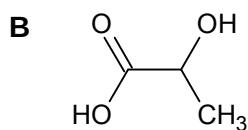
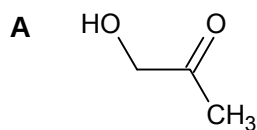
C



D

- 26 Propene reacts with aqueous bromine. The product was collected and subsequently reacted with hot potassium manganate(VII), followed by hot sodium hydroxide.

Which of this is **not** a possible correct structure of the final product?



- 27 Which of these statements is incorrect for a sheet of graphene?

- A** It is able to conduct electricity.
- B** It has a giant molecular structure.
- C** It is a nanoparticle.
- D** It is a planar molecule.

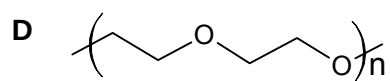
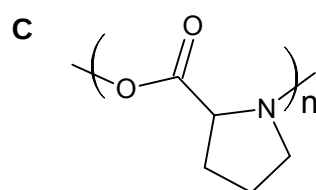
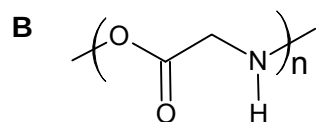
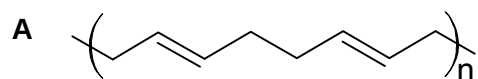
- 28 Which of these statements are correct?

- A** Nanomaterials have one of its dimensions within 10 nm to 100 nm.
- B** Lizards are able to climb on wet surfaces due to hydrogen bonding.
- C** The vivid pink colour of flamingo feathers are due to nanostructures.
- D** Picoscopic (10^{-12}) volcanic dust pose no threat when inhaled.

- 29 Which of these statements are inconsistent with the properties of a polyamide shirt?

- A** It is able to absorb water well.
- B** It is prone to creasing.
- C** It is a condensation polymer.
- D** It is resistant to basic hydrolysis.

30 Which of these shows an example of an addition polymer?



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